

**Course Code: 1BMACS301****Credits: 3****SEE: 50% Marks****SEE Hours: 3 Hrs****Course: Applied Mathematics-III for CSE Stream****L:T:P:S 28:28:0:34****CIE: 50% Marks****Max. Marks: 100**

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prerequisites if any</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Learning objectives</b>  | <p>The goal of the course Probability and Statistics is to:</p> <ol style="list-style-type: none"> <li>1. To introduce the concept of random variables, probability distributions, specific discrete and continuous distributions with practical application in Computer Science Engineering and social life situations.</li> <li>2. To provide the principles of statistical inferences and the basics of hypothesis testing with emphasis on some commonly encountered hypotheses.</li> <li>3. To determine whether an input has a statistically significant effect on the system's response through ANOVA testing.</li> </ol> |

**Course Outcomes:**

*On the successful completion of the course, the student will be able to*

| Course Outcomes |                                                                                                                                                             | Bloom's level                 |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| CO1             | Explain the basic concepts of probability, random variables, probability distribution.                                                                      | Understand,<br>Apply, Analyze |
| CO2             | Apply suitable probability distribution models for the given scenario and solve problems on discrete-time Markov chain and n-step transition probabilities. |                               |
| CO3             | Use statistical methodology and tools in the engineering problem-solving process.                                                                           |                               |
| CO4             | Compute the confidence intervals for the mean of the population and Apply the ANOVA test related to engineering problems.                                   |                               |

**Mapping with POs and PSOs:**

| COs | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |  | PSO1 | PSO2 | PSO3 | PSO4 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|--|------|------|------|------|
| CO1 | 3   | 2   | -   | -   | -   | -   | -   | -   | -   | -    | -    | 1    |  | -    | -    | -    | -    |
| CO2 | 3   | 2   | -   | -   | -   | -   | -   | -   | -   | -    | -    | 1    |  | -    | -    | -    | -    |
| CO3 | 3   | 2   | -   | -   | -   | -   | -   | -   | -   | -    | -    | 1    |  | -    | -    | -    | -    |
| CO4 | 3   | 2   | -   | -   | -   | -   | -   | -   | -   | -    | -    | 1    |  | -    | -    | -    | -    |

**Strong: 3****Medium: 2****Low: 1**

**Course Content**

|                                         | <b>Module – 1 Probability Distributions</b>                                                                                                       | <b>No. of Lecture Hours</b> | <b>No. of Tutorial Hours</b> | <b>Self-Learning Hours</b> |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------------|----------------------------|
| 1.1                                     | Review of basic probability theory.                                                                                                               | 1                           |                              |                            |
| 1.2                                     | Random variables (discrete and continuous), probability mass and density functions. Mathematical expectation, mean and variance.                  | 1                           | 1                            |                            |
| 1.3                                     | Binomial, Poisson,                                                                                                                                | 2                           |                              |                            |
| 1.4                                     | Exponential and Normal distributions- problems.                                                                                                   | 2                           | 1                            |                            |
|                                         | <b>Module – 2 Joint probability distribution &amp; Markov Chain</b>                                                                               |                             |                              |                            |
| 2.1                                     | <b>Joint probability distribution:</b> Joint Probability distribution for two discrete random variables, expectation, covariance and correlation. | 2                           | 1                            |                            |
| 2.2                                     | <b>Markov Chain:</b> Introduction to Stochastic Process, Probability Vectors, Stochastic matrices, Regular stochastic matrices.                   | 2                           | 1                            |                            |
| 2.3                                     | Markov chains, Higher transition probabilities, Stationary distribution of Regular Markov chains and absorbing states.                            | 2                           | 1                            |                            |
|                                         | <b>Module – 3 Statistical Inference 1</b>                                                                                                         |                             |                              |                            |
| 3.1                                     | Introduction, sampling distribution, standard error, testing of hypothesis                                                                        | 2                           | 1                            |                            |
| 3.2                                     | Levels of significance, test of significances, confidence limits.                                                                                 | 2                           |                              |                            |
| 3.3                                     | Simple sampling of attributes, test of significance for large samples                                                                             | 1                           | 1                            |                            |
| 3.4                                     | Comparison of large samples.                                                                                                                      | 1                           | 1                            |                            |
|                                         | <b>Module – 4 Statistical Inference 2</b>                                                                                                         |                             |                              |                            |
| 4.1                                     | Sampling variables, confidence limit for mean. Test of Significance for means of two small samples.                                               | 2                           | 1                            |                            |
| 4.2                                     | Students 't' distribution.                                                                                                                        | 1                           | 1                            |                            |
| 4.3                                     | Chi-square distribution as a test of goodness of fit.                                                                                             | 1                           | 1                            |                            |
| 4.4                                     | F-Distribution.                                                                                                                                   | 1                           |                              |                            |
|                                         | <b>Module – 5 Design of Experiments &amp; ANOVA</b>                                                                                               |                             |                              |                            |
| 5.1                                     | Principles of experimentation in design, Analysis of completely randomized design                                                                 | 2                           | 1                            |                            |
| 5.2                                     | Randomized block design. The ANOVA Technique,                                                                                                     | 1                           | 1                            |                            |
| 5.3                                     | Basic Principle of ANOVA, One-way ANOVA,                                                                                                          | 1                           |                              |                            |
| 5.4                                     | Two-way ANOVA, Latin-square Design, and Analysis of Co-Variance.                                                                                  | 1                           | 1                            |                            |
| <b>Total No. of Lecture Hours</b>       |                                                                                                                                                   | 28                          |                              |                            |
| <b>Total No. of Tutorial Hours</b>      |                                                                                                                                                   |                             | 14                           |                            |
| <b>Total No. of Self learning Hours</b> |                                                                                                                                                   |                             |                              | 34                         |

**Textbooks:**

1. **Ronald E. Walpole, Raymond H Myers, Sharon L Myers & Keying Ye** “Probability & Statistics for Engineers & Scientists”, Pearson Education, 9<sup>th</sup> edition, 2017.
2. **Peter Bruce, Andrew Bruce & Peter Gedeck** “Practical Statistics for DataScientists” O’Reilly Media, Inc., 2<sup>nd</sup> edition 2020.

**Reference Books: (Name of the author/Title of the Book/ Name of the publisher/Edition and year)**

1. **Erwin Kreyszig**, “Advanced Engineering Mathematics”, John Wiley & Sons, 9<sup>th</sup> Edition, 2006.
2. **B. S. Grewal** “Higher Engineering Mathematics”, Khanna publishers, 44<sup>th</sup> Ed., 2021.
3. **G Haribaskaran** “Probability, Queuing Theory & Reliability Engineering”, Laxmi Publication, Latest Edition, 2006
4. **Irwin Miller & Marylees Miller**, John E. Freund’s “Mathematical Statistics with Applications” Pearson. Dorling Kindersley Pvt. Ltd. India, 8<sup>th</sup> edition, 2014.
5. **S C Gupta and V K Kapoor**, “Fundamentals of Mathematical Statistics”, S Chand and Company, Latest edition.
6. **Robert V. Hogg, Joseph W. McKean & Allen T. Craig**, “Introduction to Mathematical Statistics”, Pearson Education 7<sup>th</sup> edition, 2013.
7. **Jim Pitman**, Probability, Springer-Verlag, 1993.
8. **Sheldon M. Ross**, “Introduction to Probability Models” 11<sup>th</sup> edition. Elsevier, 2014.
9. **A. M. Yaglom and I. M. Yaglom**, “Probability and Information”. D. Reidel Publishing Company. Distributed by Hindustan Publishing Corporation (India) Delhi, 1983.
10. **P. G. Hoel, S. C. Port and C. J. Stone**, “Introduction to Probability Theory”, Universal Book Stall, (Reprint), 2003.
11. **S. Ross**, “A First Course in Probability”, Pearson Education India, 6<sup>th</sup> Ed., 2002.
12. **W. Feller**, “An Introduction to Probability Theory and its Applications”, Vol. 1, Wiley, 3rd